

Ulric Aird

Foundation Math

Take Home Assignment 1

1. $a. -1 \times 0 = 0$

c. $6 \div 0 = \text{undefined}$

b. $\frac{1}{3} \times \frac{1}{0} = \frac{1}{0} = \text{undefined}$

d. $0/15 = 0$

2. Using Prime Factorisation

$$30 \text{ mins} = 2 \times 3 \times 5$$

$$45 \text{ mins} = 3^2 \times 5$$

$$50 \text{ mins} = 2 \times 5^2$$

$$\begin{aligned} \text{LCM}(30, 45, 50) &= 2 \times 3^2 \times 5^2 \\ &= 2 \times 9 \times 25 = 450 \text{ mins} \end{aligned}$$

$$\frac{450^{\text{mins}}}{60} = 7 \text{ hrs } 30 \text{ mins.}$$

$$7:30 \text{ am} + 7 \text{ hrs } 30 \text{ mins} = 15:00 \text{ or } 3:00 \text{ pm.}$$

$$\begin{aligned} 3a. & 9 \times (-1) \div 9 + 10 \times (-2) \\ &= -9 \div 9 - 20 \\ &= -1 - 20 \end{aligned}$$

$$\text{Ans} = -21$$

$$\begin{aligned}
 b. & 2^3 - 4[3 - 12 \div (-3)] + 10 \times 3 \\
 & 8 - 4[3 + 4] + 30 \\
 & 8 - 4(7) + 30 \\
 & 8 - 28 + 30 \\
 & \text{Ans} = 10
 \end{aligned}$$

$$\begin{aligned}
 4.a. & (2\frac{1}{50} - \frac{23}{25}) \div (\frac{1}{6} + \frac{3}{5}) \\
 & = (\frac{101}{50} - \frac{23 \times 2}{25}) \div (\frac{1}{6} + \frac{3}{5}) \\
 & \quad \frac{101 - 46}{50} \div \frac{5 + 18}{30} \\
 & = \frac{55}{50} \div \frac{23}{30} = \frac{55}{50} \times \frac{30}{23} \\
 & = \frac{33}{23} \\
 & \text{Ans} = 1.434782609
 \end{aligned}$$

$$b. 3dp = 1.435$$

$$5a. 24\% \text{ is } 6. 100\% = ?$$

$$\begin{aligned}
 \frac{6}{24} \times 100 &= \frac{100}{4} = 25 \\
 \text{Ans} &= 25
 \end{aligned}$$

$$b. \frac{235}{750} = 0.313 \quad \text{Ans} = 31.3\%$$

6a. ~~1.44~~ 1.42 Billion = 1.42×10^9

~~1.17~~ 1.17,000 $\rightarrow 1.17 \times 10^5$

b.
$$\frac{1.42 \times 10^9}{1.17 \times 10^5} = \frac{1.42}{1.17} \times 10^{9-5}$$

$$= 1.213675214 \times 10^4$$

 Ans = \$12,136.75

7a. $SI = \frac{\$70,000 \times 6.5 \times 5}{100}$

$SI = \$22,750$

Total = $\$70,000 + \$22,750 = \$92,750$

b. $CI = \$70,000(1 + 6\%)^5$
 $= \$70,000(1 + 0.06)^5$
 $= 70,000(1.06)^5$
 $= 70,000(1.338225578)$
 Loan = \$93,675.79

$CI = 93,675.75 - \$70,000$

$CI = \$23,675.75$

The simple interest loan is the better loan arrangement.

Q. a. 536 pupils
368 are athletes

Prime Factorisation

$$\begin{array}{l} 368 \\ \wedge \\ 2 \quad 184 \\ \wedge \\ 2 \quad 92 \\ \wedge \\ 2 \quad 46 \\ \wedge \\ 2 \quad 23 \\ = 2^4 \times 23 \end{array} \qquad \begin{array}{l} 536 \\ \wedge \\ 2 \quad 268 \\ \wedge \\ 2 \quad 134 \\ \wedge \\ 2 \quad 67 \\ = 2^3 \times 67 \end{array}$$

Finding HCF of 368 and 536
 $= 2^3 = 8$

~~Dividing~~ Divide the 2 number by 8

$$\frac{368}{8} = 46, \quad \frac{536}{8} = 67$$

Ans = $46/67$ or $46:67$

$$\text{b. Ratio} = 25:20:10 = 55$$

Ans $\rightarrow 8250$

$$8250 \div 10 = 825$$

$$= 825 \times 55$$

$$\text{Ans} = 45375 \text{ ft}^2$$

$$\text{c. Ratio} = 2:4:5 = 11$$

$$\text{Rick and Ann} = 4:2$$

$\downarrow$
receives \$220.40 more

$$4-2 = 2$$

$$\rightarrow \$220.40$$

$$220.40 \div 2 = \$110.20$$

$$\text{Betty's share} = \$110.20 \times 5$$

$$= \$551$$

$$\text{9a. } \$16.50 \times 40 \text{ hrs}$$

$$= \$740 \times 2 \text{ wks}$$

$$\text{Ans} = \$1480$$

$$\text{b. } \$16.50 \times 3 \times 1.5 = \$83.25$$

$$\$18.50 \times 8 \times 2 = \$296$$

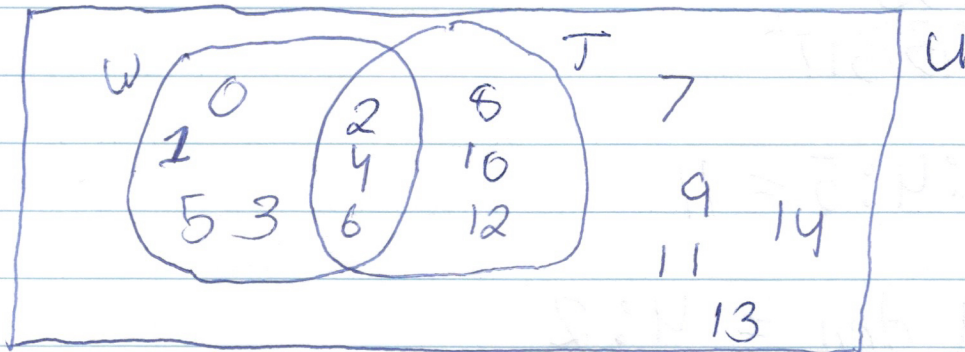
$$\$83.25 + \$296$$

$$\text{Ans} = \$379.25$$

c. $\$1480 + 379.25$
 Gross Wage = $\$1859.25$

10.

a.



b. $A \cap B = \{2, 4, 6\}$

c. $n(A \cup B) = \{0, 1, 2, 3, 4, 5, 6, 8, 10, 12\}$
 $= 10$ elements

d. How ~~etc~~ subsets can be formed from set A

~~$A = \{0, 1, 2, 3, 5, 6\}$~~ $A = \{0, 1, 2, 3, 4, 5, 6\}$
 ~~$n = 6$~~ $n = 7$ elements
 $= 2^n = 2^7$
 Ans = 128 subsets.